

Containerization_and_DevOps

SonarQube Code Quality Analysis using Docker, Maven, and Jenkins

Objective

To set up SonarQube using Docker, analyze a Java application for bugs, vulnerabilities, and code smells, and integrate it with Jenkins for CI/CD-based quality checks.

Tools & Technologies Used

- Docker & Docker Compose
- SonarQube (LTS Community Edition)
- PostgreSQL
- Maven
- Jenkins
- Java (Sample Application)

Step 1: Setup SonarQube Environment

Create Docker Network

```
docker network create sonarqube-lab
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ docker network create sonarqube-lab  
fa7961ac9251363819222ba109ab6c3ed5972f0f200344c8c5e7eb299ad59933
```

Start PostgreSQL Database

```
docker run -d \  
  --name sonar-db \  
  postgres
```

```
--network sonarqube-lab \
-e POSTGRES_USER=sonar \
-e POSTGRES_PASSWORD=sonar \
-e POSTGRES_DB=sonarqube \
-v sonar-db-data:/var/lib/postgresql/data \
postgres:13
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ docker run -d \
--name sonar-db \
--network sonarqube-lab \
-e POSTGRES_USER=sonar \
-e POSTGRES_PASSWORD=sonar \
-e POSTGRES_DB=sonarqube \
-v sonar-db-data:/var/lib/postgresql/data \
postgres:13
Unable to find image 'postgres:13' locally
13: Pulling from library/postgres
d7ecded7702a: Pull complete
bad194149a82: Pull complete
fcfd49225910: Pull complete
ba168ff08d42: Pull complete
5f1cded65589: Pull complete
947bdcac0ea3: Pull complete
4d2c7a7b3dba: Pull complete
097f34b018c0: Pull complete
8cb000c698bc: Pull complete
23a39b1003e9: Pull complete
7d0650bb210f: Pull complete
21f7a31efcb7: Pull complete
ae52c8ce1a35: Pull complete
80e8d95fdf53: Pull complete
c7bebc4b5ec1: Download complete
d932a4a7df6c: Download complete
Digest: sha256:4689940c683801b4ab839ab3b0a0a3555a5fe425371422310944e89eca7d8068
Status: Downloaded newer image for postgres:13
0cd5dc7ba9767be7c5185a2aff9ce7efc7c0cd4a1880de50338cf3aaf6f1c11a
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ |
```

Start SonarQube Server

```
docker run -d \
--name sonarqube \
--network sonarqube-lab \
-p 9000:9000 \
-e SONAR_JDBC_URL=jdbc:postgresql://sonar-db:5432/sonarqube \
-e SONAR_JDBC_USERNAME=sonar \
-e SONAR_JDBC_PASSWORD=sonar \
-v sonar-data:/opt/sonarqube/data \
-v sonar-extensions:/opt/sonarqube/extensions \
sonarqube:lts-community
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ docker run -d \
--name sonarqube \
--network sonarqube-lab \
-p 9000:9000 \
-e SONAR_JDBC_URL=jdbc:postgresql://sonar-db:5432/sonarqube \
-e SONAR_JDBC_USERNAME=sonar \
-e SONAR_JDBC_PASSWORD=sonar \
-v sonar-data:/opt/sonarqube/data \
-v sonar-extensions:/opt/sonarqube/extensions \
sonarqube:lts-community
Unable to find image 'sonarqube:lts-community' locally
lts-community: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
1df735f481ad: Pull complete
c7ad1fe61e07: Pull complete
eb27e3a98da1: Pull complete
e24a8b9e652f: Pull complete
f3929ce9ef98: Pull complete
5d5a1fad7028: Pull complete
60d98d907669: Pull complete
ce2bd68be4fc: Download complete
600f318704dc: Download complete
Digest: sha256:f709975ab31d2d08f5a3ae2dc73a31ee011afc8cf28845082c17c55d45df9df5
Status: Downloaded newer image for sonarqube:lts-community
0d40c946078334f6b91ae821044dc13b9686c8cfc3691dbfb03e0f6c51a1b568
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ |
```

Verify Logs

`docker logs -f sonarqube`

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ docker logs -f sonarqube
2026.04.18 03:13:20 INFO app[[[o.s.a.AppFileSystem] Cleaning or creating temp directory /opt/sonarqube/temp
2026.04.18 03:13:20 INFO app[[[o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:45103]
2026.04.18 03:13:20 INFO app[[[o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [/opt/sonarqube/elasticsearch]: /opt/sonarqube/elasticsearch/b
in/elasticsearch
2026.04.18 03:13:20 INFO app[[[o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
2026.04.18 03:13:22 INFO es[[[o.e.n.Node] version[7.17.15], pid[29], build[default/tar/0b8ecfb4378335f4689c4223d1f115f16bef3ba/2023-11-10T22:03:46.9873990
16Z], OS[Linux/6.6.87.2-microsoft-standard-WSL2/amd64], JVM[Eclipse Adoptium/OpenJDK 64-Bit Server VM/17.0.16/17.0.16+8]
2026.04.18 03:13:22 INFO es[[[o.e.n.Node] JVM home [/opt/java/openjdk]
2026.04.18 03:13:22 INFO es[[[o.e.n.Node] JVM arguments [-XX:+UseG1GC, -Djava.io.tmpdir=/opt/sonarqube/temp, -XX:ErrorFile=/opt/sonarqube/logs/es_hs_err_pi
d%p.log, -Des.networkaddress.cache.ttl=60, -Des.networkaddress.cache.negative.ttl=10, -XX:+AlwaysPreTouch, -Xss1m, -Djava.awt.headless=true, -Dfile.encoding
=UTF-8, -Djna.nosys=true, -Djna.tmpdir=/opt/sonarqube/temp, -XX:-OmitStackTraceInFastThrow, -Dio.netty.noUnsafe=true, -Dio.netty.noKeySetOptimization=true,
-Dio.netty.recycler.maxCapacityPerThread=0, -Dio.netty allocator.numDirectArenas=0, -Dlog4j.shutdownHookEnabled=false, -Dlog4j2.disable.jmx=true, -Dlog4j2.f
ormatMsgNoLookups=true, -Djava.locale.providers=COMPAT, -Dcom.redhat.fips=false, -Des.enforce.bootstrap.checks=true, -Xmx512m, -Xms512m, -XX:MaxDirectMemory
Size=256m, -XX:+HeapDumpOnOutOfMemoryError, -Des.path.home=/opt/sonarqube/elasticsearch, -Des.path.conf=/opt/sonarqube/temp/conf/es, -Des.distribution.flavo
r=default, -Des.distribution.type=tar, -Des.bundled_jdk=false]
2026.04.18 03:13:22 INFO es[[[o.e.p.PluginsService] loaded module [analysis-common]
2026.04.18 03:13:22 INFO es[[[o.e.p.PluginsService] loaded module [lang-painless]
2026.04.18 03:13:22 INFO es[[[o.e.p.PluginsService] loaded module [parent-join]
2026.04.18 03:13:22 INFO es[[[o.e.p.PluginsService] loaded module [reindex]
2026.04.18 03:13:22 INFO es[[[o.e.p.PluginsService] loaded module [transport-netty4]
2026.04.18 03:13:22 INFO es[[[o.e.p.PluginsService] no plugins loaded
2026.04.18 03:13:22 INFO es[[[o.e.e.NodeEnvironment] using [1] data paths, mounts [[/opt/sonarqube/data (/dev/sdf)], net usable_space [939.8gb], net total
_space [1006.8gb], types [ext4]
2026.04.18 03:13:22 INFO es[[[o.e.e.NodeEnvironment] heap size [512mb], compressed ordinary object pointers [true]
2026.04.18 03:13:22 INFO es[[[o.e.n.Node] node name [sonarqube], node ID [piDVpyRsQxGKaDl7Kjmiuw], cluster name [sonarqube], roles [data_frozen, master, re
mote_cluster_client, data, data_content, data_hot, data_warm, data_cold, ingest]
2026.04.18 03:13:22 WARN es[[[o.e.t.ThreadPool] timer thread slept for [26.3s/26399ms] on absolute clock which is above the warn threshold of [5000ms]
2026.04.18 03:13:22 WARN es[[[o.e.t.ThreadPool] absolute clock went backwards by [25.9s/25996ms] while timer thread was sleeping
2026.04.18 03:13:25 INFO es[[[o.e.t.NettyAllocator] creating NettyAllocator with the following configs: [name=unpooled, suggested_max_allocation_size=256kb
, factors={es.unsafe.use_unpooled_allocator=null, g1gc_enabled=true, g1gc_region_size=1mb, heap_size=512mb}]
2026.04.18 03:13:25 INFO es[[[o.e.i.r.RecoverySettings] using rate limit [40mb] with [default=40mb, read=0b, write=0b, max=0b]
2026.04.18 03:13:25 INFO es[[[o.e.d.DiscoveryModule] using discovery type [zen] and seed hosts providers [settings]
2026.04.18 03:13:25 INFO es[[[o.e.g.DanglingIndicesState] gateway.auto_import_dangling_indices is disabled, dangling indices will not be automatically dete
cted or imported and must be managed manually
2026.04.18 03:13:25 INFO es[[[o.e.n.Node] initialized
2026.04.18 03:13:25 INFO es[[[o.e.n.Node] starting ...
2026.04.18 03:13:25 INFO es[[[o.e.t.TransportService] publish_address [127.0.0.1:45103], bound_addresses [127.0.0.1:45103]
2026.04.18 03:13:25 INFO es[[[o.e.b.BootstrapChecks] explicitly enforcing bootstrap checks
2026.04.18 03:13:25 INFO es[[[o.e.c.c.Coordinator] setting initial configuration to VotingConfiguration{piDVpyRsQxGKaDl7Kjmiuw}]
```

Access SonarQube

- URL: `http://localhost:9000`
- Default Login: `admin / admin`

The first screenshot shows the SonarQube maintenance page at `localhost:9000/maintenance?return_to=%2F`. It displays a message: "SonarQube is up" and "All systems operational." with a [Home](#) link.

The second screenshot shows the login page at `localhost:9000/sessions/new?return_to=%2F`. It has a title "Log in to SonarQube" and fields for "Login" and "Password". There are "Log in" and "Cancel" buttons.

The third screenshot shows the project creation page at `localhost:9000/projects/create`. It has a navigation bar with links: sonarqube, Projects, Issues, Rules, Quality Profiles, Quality Gates, and Administration. A search bar is present. The main content asks "How do you want to create your project?" and provides options to create a project from various DevOps platforms (Azure DevOps, Bitbucket Server, Bitbucket Cloud, GitHub, GitLab) or manually. Each option includes a "Set up global configuration" link. At the bottom, it says "SonarQube™ technology is powered by SonarSource SA".



Alternative: Docker Compose Setup

```
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    restart: unless-stopped
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    restart: unless-stopped
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    depends_on:
      - sonar-db
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
  sonar-extensions:

networks:
  sonarqube-lab:
    driver: bridge
```

Run:

```
docker compose up -d
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab$ docker compose up -d
WARN[0000] /mnt/c/Users/ASUS/sonarqube-lab/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion
[+] up 2/2
✔Container sonar-db Created 0.1s
✔Container sonarqube Created 0.2s
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab$ |
```



Step 2: Create Sample Java Application

Create Project Structure

```
mkdir -p sample-java-app/src/main/java/com/example
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab$ mkdir -p sample-java-app/src/main/java/com/example
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab$ cat > sample-java-app/src/main/java/com/example/Calculator.java << 'EOF'
```

Add Java Code (with issues)

(Create Calculator.java as provided in experiment)

```

aaroji26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab$ cat > sample-java-app/src/main/java/com/example/Calculator.java << 'EOF'
> package com.example;

import java.util.ArrayList;
import java.util.List;

public class Calculator {

    // Bug: Division by zero not handled
    public int divide(int a, int b) {
        return a / b; // Bug: Potential division by zero
    }

    // Code Smell: Unused variable
    public int add(int a, int b) {
        int result = a + b;
        int unused = 100; // Code smell: Unused variable
        return result;
    }

    // Vulnerability: SQL injection risk
    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId; // Vulnerability: SQL injection
        return query;
    }

    // Code Smell: Duplicate code
    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    // Duplicate code (same as multiply method)
    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }
} // Empty catch block - swallowing exception if name is null
aaroji26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab$ cat sample-java-app/src/main/java/com/example/Calculator.java
package com.example;

import java.util.ArrayList;
import java.util.List;

public class Calculator {

```

```

j
aaruhi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab$ cat > sample-java-app/pom.xml << 'EOF'
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>sample-app</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <maven.compiler.source>11</maven.compiler.source>
    <maven.compiler.target>11</maven.compiler.target>
    <sonar.projectKey>sample-java-app</sonar.projectKey>
    <sonar.organization>my-org</sonar.organization>
    <sonar.host.url>http://localhost:9000</sonar.host.url>
    <sonar.login>your-sonar-token</sonar.login>
  </properties>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.sonarsource.scanner.maven</groupId>
        <artifactId>sonar-maven-plugin</artifactId>
        <version>3.9.1.2184</version>
      </plugin>
    </plugins>
  </build>
</project>
EOF
```

Step 3: Maven Configuration

Create pom.xml

(Use provided pom.xml)

Build Project

```
cd sample-java-app
mvn clean install
```

```

aaroHi26@LAPTOP-4MMNSFCL:/mnt/c/Users/ASUS/sonarqube-lab$ cd sample-java-app
mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] < com.example:sample-app >
[INFO] Building sample-app 1.0-SNAPSHOT
[INFO] [ jar ]
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.pom (10 kB at 14 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/24/maven-plugins-24.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/24/maven-plugins-24.pom (11 kB at 140 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.jar (43 kB at 275 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/2.4/maven-install-plugin-2.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/2.4/maven-install-plugin-2.4.pom (6.4 kB at 60 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/2.4/maven-install-plugin-2.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/2.4/maven-install-plugin-2.4.jar (27 kB at 270 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/junit/junit4.13.2/junit-4.13.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit4.13.2/junit-4.13.2.pom (27 kB at 311 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.pom (766 B at 29 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-parent/1.3/hamcrest-parent-1.3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-parent/1.3/hamcrest-parent-1.3.pom (2.0 kB at 18 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/junit/junit4.13.2/junit-4.13.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit4.13.2/junit-4.13.2.jar (385 kB)
Downloading from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.jar (45 kB)
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ sample-app ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ sample-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /mnt/c/Users/ASUS/sonarqube-lab/sample-java-app/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ sample-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.5.1/plexus-utils-3.5.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.5.1/plexus-utils-3.5.1.pom (640 kB)
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /mnt/c/Users/ASUS/sonarqube-lab/sample-java-app/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ sample-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /mnt/c/Users/ASUS/sonarqube-lab/sample-java-app/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ sample-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ sample-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-jar-plugin:2.4:jar (default-jar) @ sample-app ---
[INFO] Building jar: /mnt/c/Users/ASUS/sonarqube-lab/sample-java-app/target/sample-app-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ sample-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.5/plexus-utils-3.0.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.5/plexus-utils-3.0.5.pom (2.5 kB at 43 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/3.1/plexus-3.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/3.1/plexus-3.1.pom (19 kB at 345 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.pom (1.1 kB at 22 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-components/1.1.7/plexus-components-1.1.7.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-components/1.1.7/plexus-components-1.1.7.pom (5.0 kB at 77 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.8/plexus-1.0.8.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.8/plexus-1.0.8.pom (7.2 kB at 99 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-container-default/1.0-alpha-8/plexus-container-default-1.0-alpha-8.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-container-default/1.0-alpha-8/plexus-container-default-1.0-alpha-8.pom (7.3 kB at 119 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.5/plexus-utils-3.0.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.5/plexus-utils-3.0.5.jar (12 kB at 198 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar (230 kB at 3.5 MB/s)
[INFO] Installing /mnt/c/Users/ASUS/sonarqube-lab/sample-java-app/target/sample-app-1.0-SNAPSHOT.jar to /home/aaroHi26/.m2/repository/com/example/sample-app/1.0-SNAPSHOT/sample-app-1.0-SNAPSHOT.jar
[INFO] Installing /mnt/c/Users/ASUS/sonarqube-lab/sample-java-app/pom.xml to /home/aaroHi26/.m2/repository/com/example/sample-app/1.0-SNAPSHOT/sample-app-1.0-SNAPSHOT.pom
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 10.167 s
[INFO] Finished at: 2026-04-18T03:41:50Z
[INFO]
aaroHi26@LAPTOP-4MMNSFCL:/mnt/c/Users/ASUS/sonarqube-lab/sample-java-app$

```

Step 4: SonarQube Scanner Setup

Run Scanner via Docker

```

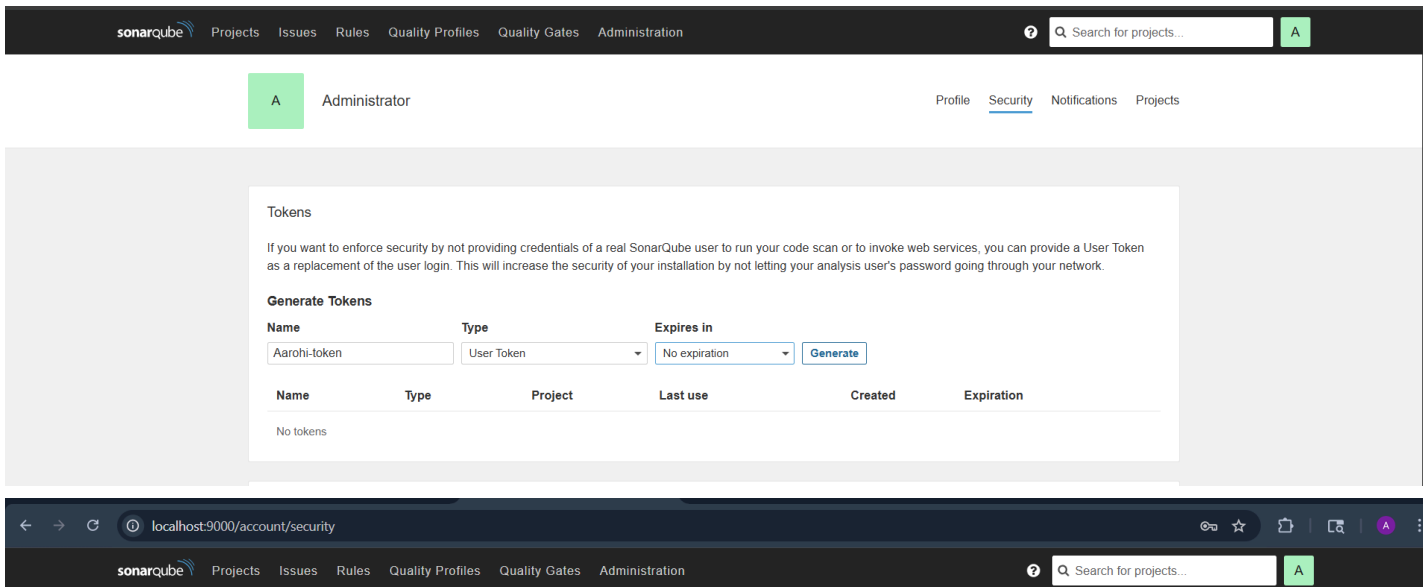
docker run -d \
  --name sonar-scanner \
  --network sonarqube-lab \
  -v $(pwd)/sample-java-app:/usr/src \
  sonarsource/sonar-scanner-cli \
  sleep infinity

```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab/sample-java-app$ docker run -d \
--name sonar-scanner \
--network sonarqube-lab \
-v $(pwd):/usr/src \
sonarsource/sonar-scanner-cli \
sleep infinity
520fa5f51b73cc2a40d42e7fcbcd7c292643ebc3b7fe26545d1474085b5565b
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/sonarqube-lab/sample-java-app$ |
```

Step 5: Generate Token

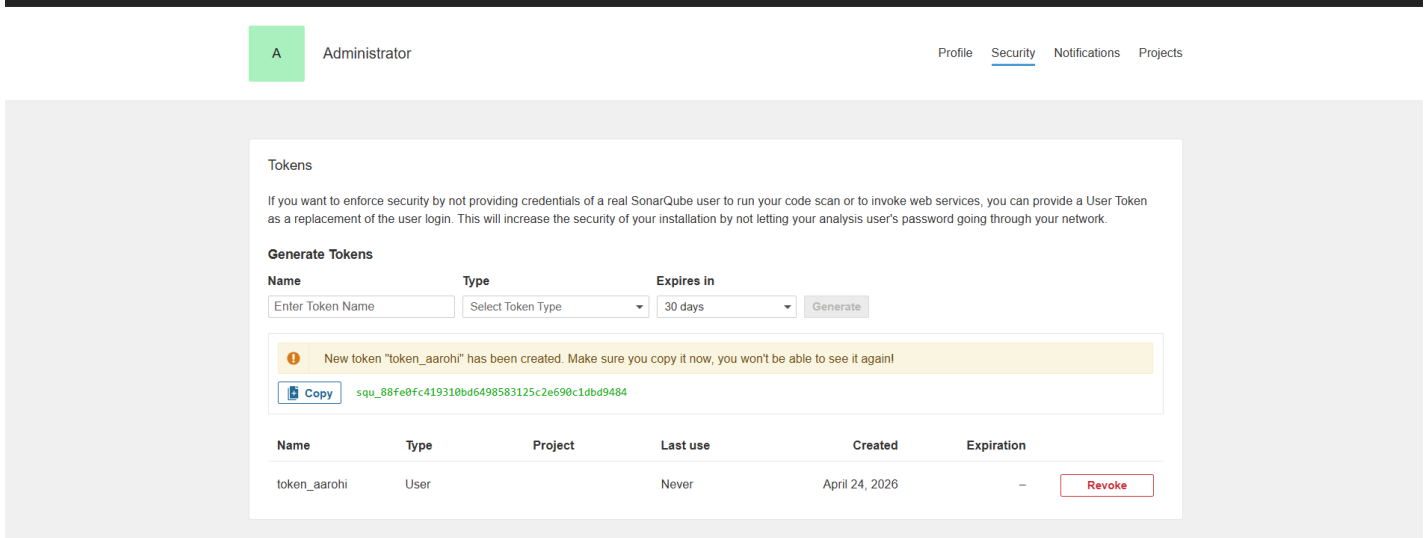
- Go to SonarQube Dashboard
- My Account → Security → Generate Token
- Copy token



The screenshot shows the SonarQube dashboard with the 'Security' tab selected. Under 'Generate Tokens', there is a form with the following fields:

Name	Type	Expires in
Aarohi-token	User Token	No expiration

A 'Generate' button is next to the 'Expires in' field. Below the form is a table with columns: Name, Type, Project, Last use, Created, and Expiration. It currently shows 'No tokens'.



The screenshot shows the SonarQube dashboard with the 'Security' tab selected. A message indicates that a new token has been created. The token value is displayed as 'squ_88fe0fc419310bd6498583125c2e690c1dbd9484'. The 'Generate Tokens' form is now disabled. Below the form is a table with columns: Name, Type, Project, Last use, Created, and Expiration. It now shows the newly created token.

Name	Type	Project	Last use	Created	Expiration
token_aarohi	User		Never	April 24, 2026	-



Step 6: Run Code Analysis

```
docker exec -e SONAR_TOKEN="your-token" \  
  sonar-scanner \  
  sonar-scanner \  
  -Dsonar.host.url=http://sonarqube:9000 \  
  -Dsonar.projectKey=sample-java-app \  
  -Dsonar.sources=/usr/src/src
```

```
aaron126@LAPTOP-4HNN5FLL:/mnt/c/Users/ASUS$ docker exec -e SONAR_TOKEN="squ_6d9bee3850071507b0e51e0ca59a3c4c82ed61/1" sonar-scanner sonar-scanner -Dsonar.host.url=http://sonarqube:9000 -Dsonar.projectKey=sample-java-app -Dsonar.sources=/usr/src/src  
05:53:10.096 INFO Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties  
05:53:10.101 INFO Project root configuration file: NONE  
05:53:10.119 INFO SonarScanner CLI 8.0.1.6346  
05:53:10.120 INFO Linux 6.6.87.2-microsoft-standard-WSL2 amd64  
05:53:15.770 ERROR Failed to query server version: Call to URL [http://sonarqube:9000/api/v2/analysis/version] failed: null  
05:53:15.771 INFO EXECUTION FAILURE  
05:53:15.772 INFO Total time: 5.681s
```



Step 7: View Results

Open:

<http://localhost:9000/dashboard?id=sample-java-app>

```

06:18:22.470 ERROR
06:18:22.470 ERROR Re-run SonarScanner CLI using the -X switch to enable full debug logging.
aaron@1248:~$ docker exec sonar-scanner sh -c 'sonar-scanner \
-Dsonar.host.url=http://host.docker.internal:9000 \
-Dsonar.projectKey=sample-java-app \
-Dsonar.sources=/usr/src/src \
-Dsonar.log.level=DEBUG \
-Dsonar.testExecutionReportPaths=/opt/sonar-scanner/conf/sonar-scanner.properties
06:13:22.570 INFO Scanner configuration file: /opt/sonar-scanner/conf/sonar-scanner.properties
06:13:22.587 INFO Project root configuration file: NONE
06:13:22.603 INFO SonarScanner CLI 8.0.1.6346
06:13:22.610 INFO Linux 6.6.87.2-microsoft-standard-WSL2 amd64
06:13:24.004 INFO Communicating with SonarQube Server 9.9.8.100196
06:13:24.527 INFO Load global settings
06:13:24.605 INFO Load global settings (done) | time=79ms
06:13:24.607 INFO Server id: 74C15348-A22e1JVU4cMHAZtvtG1U
06:13:24.614 INFO User cache: /opt/sonar-scanner/.sonar/cache
06:13:24.619 INFO Load/download plugins
06:13:24.619 INFO Load plugins index
06:13:24.681 INFO Load plugins index (done) | time=62ms
06:13:31.614 INFO Load/download plugins (done) | time=6995ms
06:13:32.124 INFO Process project properties
06:13:32.139 INFO Process project properties (done) | time=15ms
06:13:32.140 INFO Execute project builders
06:13:32.145 INFO Execute project builders (done) | time=2ms
06:13:32.161 INFO Project key: sample-java-app
06:13:32.161 INFO Base dir: /usr/src
06:13:32.161 INFO Working dir: /usr/src/.scannerwork
06:13:32.186 INFO Load project settings for component key: 'sample-java-app'
06:13:32.350 WARN SCM provider autodetection failed. Please use "sonar.scm.provider" to define SCM of your project, or disable the SCM Sensor in the project settings.
06:13:32.368 INFO Load quality profiles
06:13:32.458 INFO Load quality profiles (done) | time=90ms
06:13:32.473 INFO Load active rules
06:13:34.734 INFO Load active rules (done) | time=2261ms
06:13:34.738 INFO Load analysis cache
06:13:34.748 INFO Load analysis cache (404) | time=10ms
06:13:34.840 INFO Load project repositories
06:13:34.860 INFO Load project repositories (done) | time=20ms
06:13:34.880 INFO Indexing files
06:13:34.889 INFO Project configuration:
06:14:04.194 INFO 1 file indexed
06:14:04.195 INFO Quality profile for java: Sonar way
06:14:04.195 INFO ----- Run sensors on module sample-java-app
06:14:04.270 INFO Load metrics repository
06:14:04.313 INFO Load metrics repository (done) | time=40ms
06:13:36.880 INFO Sensor JavaSensor [java]
06:13:36.885 INFO Configured Java source version (sonar.java.source): none
06:13:36.888 INFO JavaClasspath initialization
06:13:36.889 INFO JavaClasspath initialization (done) | time=1ms
06:13:36.889 INFO JavaTestClasspath initialization
06:13:36.890 INFO JavaTestClasspath initialization (done) | time=1ms
06:13:36.896 INFO Server-side caching is enabled. The Java analyzer will not try to leverage data from a previous analysis.
06:13:36.898 INFO Using ECJ batch to parse 1 Main java source files with batch size 101 MB.
06:13:37.167 INFO Starting batch processing.
06:14:07.270 INFO The Java analyzer cannot skip unchanged files in this context. A full analysis is performed for all files.
06:13:38.906 INFO 100% analyzed
06:13:38.906 INFO Batch processing: Done.
06:13:38.908 INFO Did not optimize analysis for any files, performed a full analysis for all 1 files.
06:13:38.910 INFO No "Test" source files to scan.
06:13:38.910 INFO No "Generated" source files to scan.
06:13:38.910 INFO Sensor JavaSensor [java] (done) | time=2030ms
06:13:38.910 INFO Sensor JaCoCo XML Report Importer [jacoco]
06:13:38.914 INFO 'sonar.coverage.jacoco.xmlReportPaths' is not defined. Using default locations: target/site/jacoco/jacoco.xml,target/site/jacoco-it/jacoco.xml,build/reports/jacoco/test/jacocoTestReport.xml
06:13:38.920 INFO No report imported, no coverage information will be imported by JaCoCo XML Report Importer
06:13:38.920 INFO Sensor JaCoCo XML Report Importer [jacoco] (done) | time=10ms
06:13:38.920 INFO Sensor CSS Rules [javascript]
06:13:38.929 INFO No CSS, PHP, HTML or VueJS files are found in the project. CSS analysis is skipped.
06:13:38.929 INFO Sensor CSS Rules [javascript] (done) | time=1ms
06:13:38.929 INFO Sensor C# Project Type Information [csharp]
06:13:38.931 INFO Sensor C# Project Type Information [csharp] (done) | time=2ms
06:13:38.931 INFO Sensor C# Analysis Log [csharp]
06:13:38.991 INFO Sensor C# Analysis Log [csharp] (done) | time=60ms
06:13:38.991 INFO Sensor C# Properties [csharp]
06:13:38.991 INFO Sensor C# Properties [csharp] (done) | time=0ms
06:13:38.991 INFO Sensor SurefireSensor [java]
06:13:38.992 INFO parsing [/usr/src/target/surefire-reports]
06:13:38.995 INFO Sensor SurefireSensor [java] (done) | time=4ms
06:13:38.996 INFO Sensor HTML [web]
06:13:38.982 INFO Sensor HTML [web] (done) | time=6ms
06:13:39.003 INFO Sensor TextAndSecretsSensor [text]
06:13:39.036 INFO 1 source file to be analyzed
06:13:39.108 INFO 1/1 source file has been analyzed
06:13:39.108 INFO Sensor TextAndSecretsSensor [text] (done) | time=105ms
06:13:39.109 INFO Sensor VB.NET Project Type Information [vbnet]
06:13:39.109 INFO Sensor VB.NET Project Type Information [vbnet] (done) | time=1ms
06:13:39.109 INFO Sensor VB.NET Analysis Log [vbnet]
06:13:39.131 INFO Sensor VB.NET Analysis Log [vbnet] (done) | time=22ms
06:13:39.131 INFO Sensor VB.NET Properties [vbnet]
06:13:39.131 INFO Sensor VB.NET Properties [vbnet] (done) | time=0ms
06:13:39.131 INFO Sensor Iac Docker Sensor [iac]
06:13:39.134 INFO 0 source files to be analyzed
06:13:39.209 INFO 0/0 source files have been analyzed
06:13:39.210 INFO Sensor Iac Docker Sensor [iac] (done) | time=79ms
06:13:39.213 INFO ----- Run sensors on project
06:13:39.270 INFO Sensor Analysis Warnings Import [csharp]
06:13:39.280 INFO Sensor Analysis Warnings Import [csharp] (done) | time=1ms
06:13:39.280 INFO Sensor Zero Coverage Sensor

```

Observed Issues:

- Bugs: Division by zero, Null pointer
- Vulnerability: SQL Injection
- Code Smells:
 - Unused variables
 - Duplicate code
 - Too many parameters
 - Empty catch block

Metrics:

- Bugs: ~8
- Vulnerabilities: 1
- Code Smells: ~12
- Duplications: 2 blocks
- Test Coverage: 0%
- Technical Debt: ~2 hours

Challenges Faced & Solutions

1. SonarQube Not Starting

- Cause: Low system memory
- Solution:

```
sysctl -w vm.max_map_count=262144
```

2. Database Connection Failure

- Cause: Incorrect JDBC URL
- Solution: Ensure correct service name (sonar-db)

3. Scanner Not Connecting

- Cause: Network mismatch
- Solution: Use same Docker network

4. Token Authentication Error

- Cause: Invalid token
- Solution: Regenerate token



Conclusion

This experiment demonstrated how to:

- Deploy SonarQube using Docker
- Analyze code quality issues in a Java application
- Identify bugs, vulnerabilities, and code smells
- Integrate SonarQube with Jenkins for CI/CD
- Enforce Quality Gates before deployment

SonarQube helps improve code reliability, maintainability, and security by detecting issues early in the development lifecycle.



Future Improvements

- Add unit test coverage
- Fix detected issues
- Integrate with GitHub Actions
- Enable automated reports